

Objective:

To understand and implement advanced components in Java Swing for building interactive and organized GUI-based applications.

Remaining essential GUI components in Java Swing. This includes:

1. JTextArea
2. JCheckBox
3. JRadioButton
4. JComboBox
5. JPanel
6. Extra Important Component: JTable (for displaying tabular data)

Component 6: JTextArea (Multi-line Text Input)

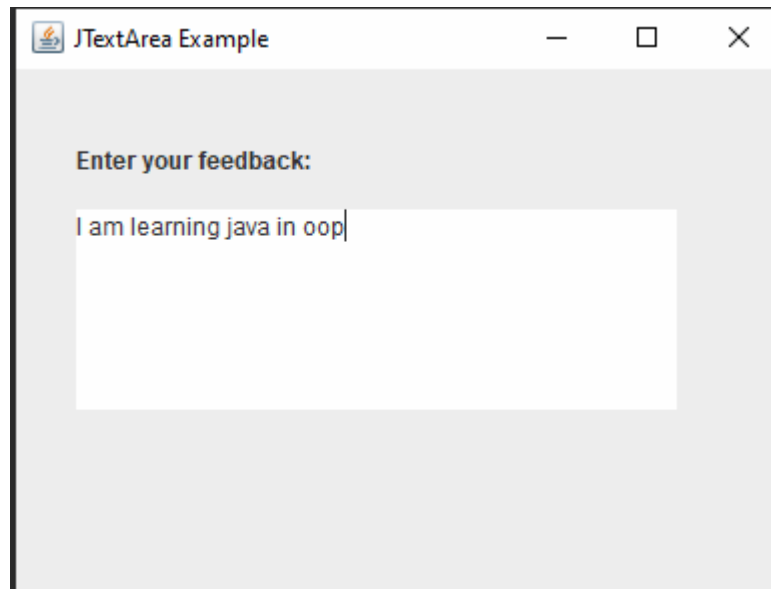
Purpose: Allows users to write long messages or feedback.

Example Use: Feedback form or note-taking section.

```
import javax.swing.*;

public class TextAreaExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("JTextArea Example");
        frame.setSize(400, 300);
        frame.setLayout(null);
        // Label to guide user
        JLabel label = new JLabel("Enter your feedback:");
        label.setBounds(30, 30, 200, 30);
        frame.add(label);
        // JTextArea allows multi-line input
        JTextArea area = new JTextArea();
        area.setBounds(30, 70, 300, 100);
        frame.add(area);

        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



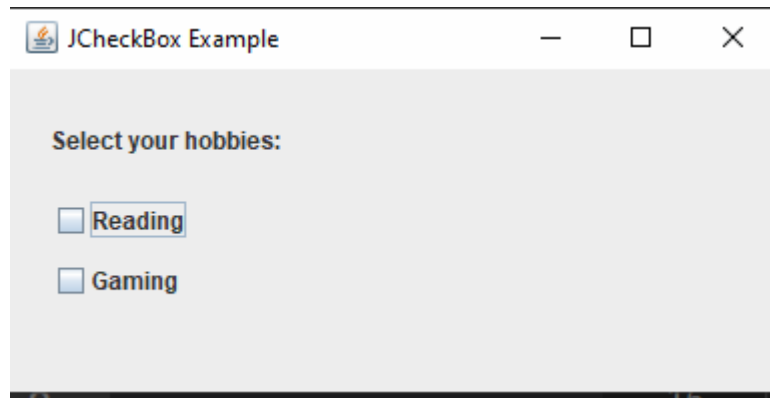
Component 7: JCheckBox (Multiple Options Selection)

Purpose: Allows users to select more than one item from options.

Example Use: What hobbies do you like?

```
import javax.swing.*;

public class CheckBoxExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("JCheckBox Example");
        frame.setSize(400, 200);
        frame.setLayout(null);
        JLabel label = new JLabel("Select your hobbies:");
        label.setBounds(20, 20, 200, 30);
        frame.add(label);
        // User can select one or more checkboxes
        JCheckBox cb1 = new JCheckBox("Reading");
        cb1.setBounds(20, 60, 100, 30);
        JCheckBox cb2 = new JCheckBox("Gaming");
        cb2.setBounds(20, 90, 100, 30);
        frame.add(cb1);
        frame.add(cb2);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



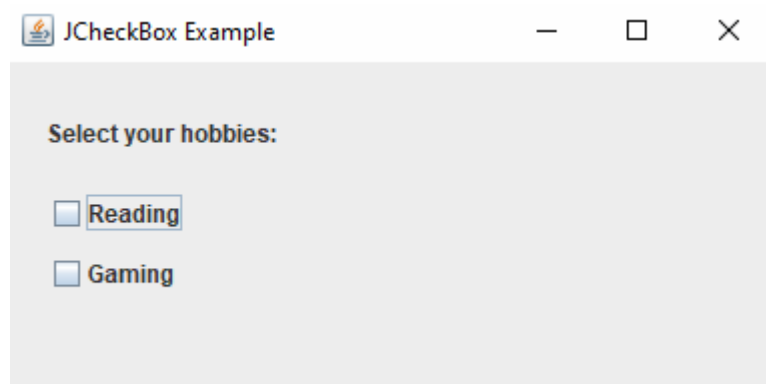
Component 8: JRadioButton (Single Option Selection)

Purpose: Only one option can be selected at a time.

Example Use: Select Gender: Male or Female.

```
import javax.swing.*;

public class RadioButtonExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("JRadioButton Example");
        frame.setSize(400, 200);
        frame.setLayout(null);
        JLabel label = new JLabel("Select your gender:");
        label.setBounds(20, 20, 200, 30);
        frame.add(label);
        JRadioButton r1 = new JRadioButton("Male");
        r1.setBounds(20, 60, 100, 30);
        JRadioButton r2 = new JRadioButton("Female");
        r2.setBounds(20, 90, 100, 30);
        // Group so only one can be selected
        ButtonGroup group = new ButtonGroup();
        group.add(r1);
        group.add(r2);
        frame.add(r1);
        frame.add(r2);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



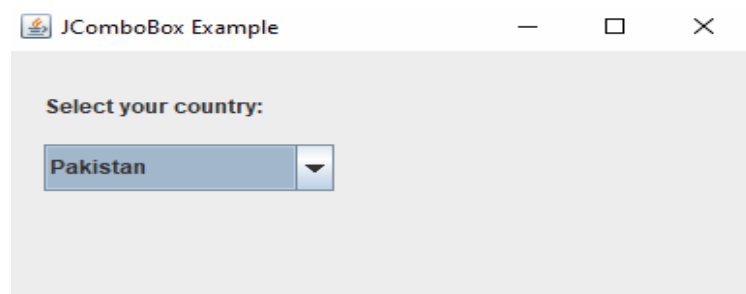
Component 9: JComboBox (Dropdown Menu)

Purpose: Provides a dropdown list for selecting one item.

Example Use: Select your country from the list.

```
import javax.swing.*.*;

public class JComboBoxExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("JComboBox Example");
        frame.setSize(400, 200);
        frame.setLayout(null);
        JLabel label = new JLabel("Select your country:");
        label.setBounds(20, 20, 200, 30);
        frame.add(label);
        // Drop-down menu with options
        String[] countries = {"Pakistan", "India", "USA", "UK"};
        JComboBox<String> comboBox = new JComboBox<>(countries);
        comboBox.setBounds(20, 60, 150, 30);
        frame.add(comboBox);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



Component 10: JPanel (Organizing Layout)

Purpose: Used to group components inside a window.

Example Use: Header section or form group inside the main frame.

```
import javax.swing.*;
import java.awt.*;

public class JPanelGroupExample {
    public static void main(String[] args) {
        // Create main frame
        JFrame frame = new JFrame("Login Form Using JPanel");
        frame.setSize(400, 350);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new BorderLayout()); // Main layout

        // ----- Header Panel -----
        JPanel headerPanel = new JPanel(); // Top part
        JLabel headerLabel = new JLabel("User Login");
        headerPanel.add(headerLabel); // Add label to panel

        // ----- Form Panel (Center) -----
        JPanel formPanel = new JPanel();
        formPanel.setLayout(new GridLayout(2, 2, 10, 10)); // 2 rows, 2 columns

        // Username row
        JLabel userLabel = new JLabel("Username:");
        JTextField userField = new JTextField();

        // Password row
        JLabel passLabel = new JLabel("Password:");
        JPasswordField passField = new JPasswordField();

        // Add form fields to form panel
        formPanel.add(userLabel);
        formPanel.add(userField);
        formPanel.add(passLabel);
        formPanel.add(passField);

        // ----- Button Panel -----
        JPanel buttonPanel = new JPanel(); // Bottom part
        JButton loginButton = new JButton("Login");
        JButton cancelButton = new JButton("Cancel");
        buttonPanel.add(loginButton);
        buttonPanel.add(cancelButton);

        // ----- Add all panels to main frame -----
        frame.add(headerPanel, BorderLayout.NORTH); // Top
        frame.add(formPanel, BorderLayout.CENTER); // Center
        frame.add(buttonPanel, BorderLayout.SOUTH); // Bottom
        frame.setVisible(true); // Show the window
    }
}
```

Bonus Component: JTable (Tabular Data Display)

Purpose: Displays data in rows and columns like Excel.

Example Use: Show student record in a table.

```
import javax.swing.*;

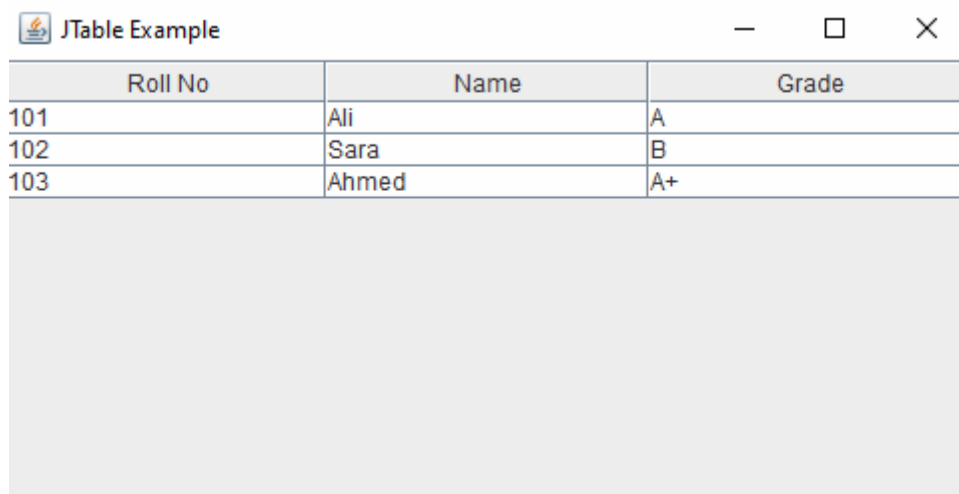
public class TableExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("JTable Example");
        frame.setSize(500, 300);

        // Sample data
        String[][] data = {
            {"101", "Ali", "A"},
            {"102", "Sara", "B"},
            {"103", "Ahmed", "A+"}
        };

        // Column headers
        String[] columns = {"Roll No", "Name", "Grade"};

        JTable table = new JTable(data, columns);
        JScrollPane sp = new JScrollPane(table); // Add scrolling if needed
        frame.add(sp);

        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



The screenshot shows a Java Swing window titled "JTable Example" with standard Windows window controls (minimize, maximize, close). Inside the window is a JTable with three columns: "Roll No", "Name", and "Grade". The table contains three rows of data: (101, Ali, A), (102, Sara, B), and (103, Ahmed, A+). Below the table is a large, empty, light gray rectangular area, likely a placeholder for additional content or a scrollable region.

Roll No	Name	Grade
101	Ali	A
102	Sara	B
103	Ahmed	A+

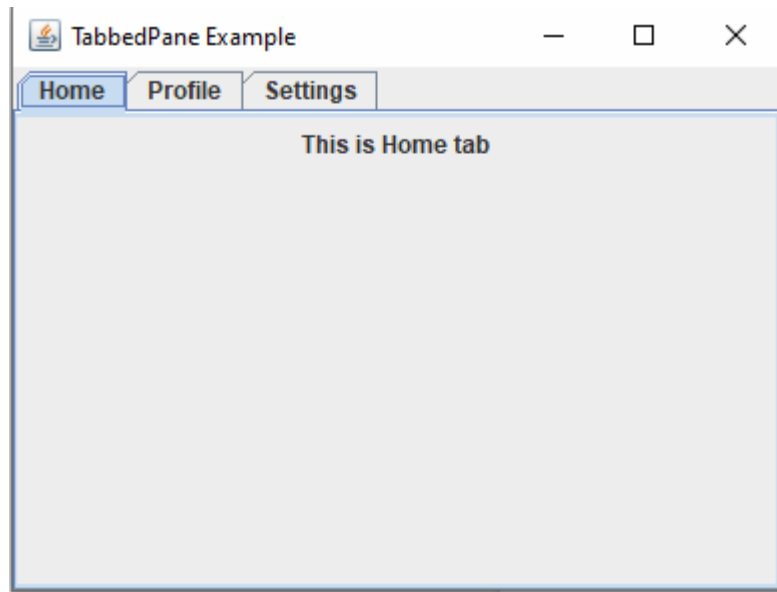
Now let us study More Interesting Things in Java GUI

1: JTabbedPane (Multiple Tabs like Browser)

Objective: Teach how to organize multiple sections in the same window using tabs.

```
import javax.swing.*;

public class TabbedPaneExample {
    public static void main(String[] args) {
        // Create the main frame
        JFrame frame = new JFrame("TabbedPane Example");
        frame.setSize(400, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        // Create a tabbed pane (like browser tabs)
        JTabbedPane tabbedPane = new JTabbedPane();
        // First Tab
        JPanel panel1 = new JPanel();
        panel1.add(new JLabel("This is Home tab"));
        tabbedPane.add("Home", panel1);
        // Second Tab
        JPanel panel2 = new JPanel();
        panel2.add(new JLabel("This is Profile tab"));
        tabbedPane.add("Profile", panel2);
        // Third Tab
        JPanel panel3 = new JPanel();
        panel3.add(new JLabel("This is Settings tab"));
        tabbedPane.add("Settings", panel3);
        // Add tabs to frame
        frame.add(tabbedPane);
        frame.setVisible(true);
    }
}
```

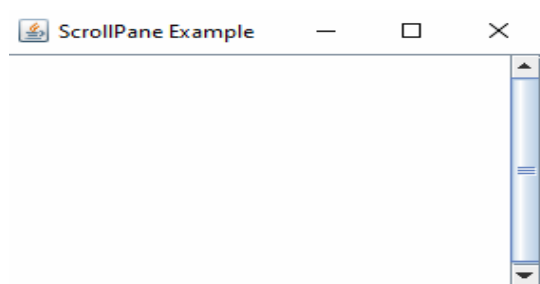


2: JScrollPane (Add Scrollbars for Long Content)

Objective: Teach how to scroll through long text using a scroll bar.

```
import javax.swing.*;

public class JScrollPaneExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("ScrollPane Example");
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        // Create a text area with 10 rows and 20 columns
        JTextArea textArea = new JTextArea(10, 20);
        // Add the text area to a scroll pane
        JScrollPane scrollPane = new JScrollPane(textArea);
        // Add the scroll pane to the frame
        frame.add(scrollPane);
        frame.setVisible(true);
    }
}
```



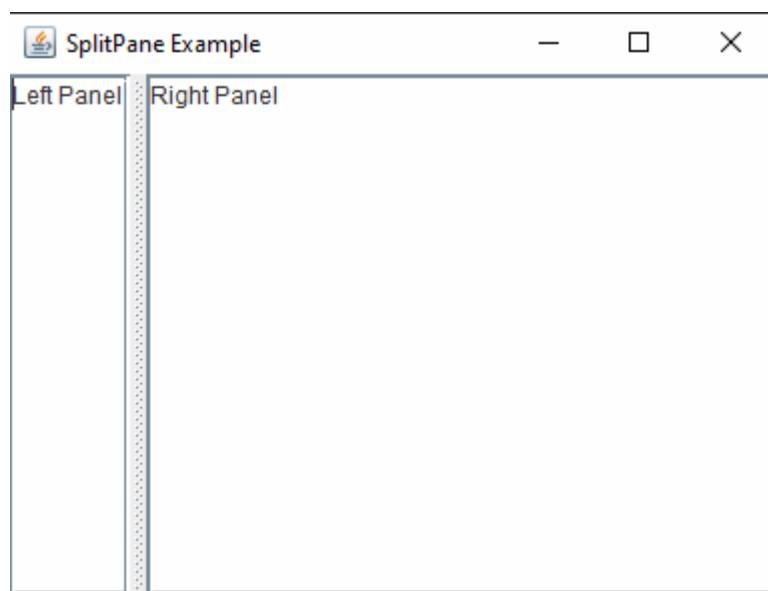
3: JSplitPane (Split Screen Layout)

Objective: Teach how to split the window into two adjustable parts.

```
import javax.swing.*.*;

public class SplitPaneExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("SplitPane Example");
        frame.setSize(400, 300);

        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        // Create two text areas for left and right side
        JTextArea leftArea = new JTextArea("Left Panel");
        JTextArea rightArea = new JTextArea("Right Panel");
        // Create a horizontal split pane (left-right)
        JSplitPane splitPane = new
JSplitPane(JSplitPane.HORIZONTAL_SPLIT,
            new JScrollPane(leftArea), new
JScrollPane(rightArea));
        // Add split pane to frame
        frame.add(splitPane);
        frame.setVisible(true);
    }
}
```



4: JMenuBar (Top Menu like File, Edit)

Objective: Create menus like File > New, Exit like in real applications.

```
import javax.swing.*;

public class MenuBarExample {
    public static void main(String[] args) {
        JFrame frame = new JFrame("MenuBar Example");
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        // Create a menu bar
        JMenuBar menuBar = new JMenuBar();
        // Create a "File" menu
        JMenu fileMenu = new JMenu("File");
        // Create menu items under File
        JMenuItem newItem = new JMenuItem("New");
        JMenuItem exitItem = new JMenuItem("Exit");
        // Add menu items to the File menu
        fileMenu.add(newItem);
        fileMenu.add(exitItem);
        // Add File menu to the menu bar
        menuBar.add(fileMenu);
        // Set the menu bar in the frame
        frame.setJMenuBar(menuBar);
        frame.setVisible(true);
    }
}
```

